

VANISHING OF HARMONIC FUNCTIONS IN PLANAR DOMAINS

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1. INTRODUCTION

Let Ω be an open and connected subset of the complex plane $\mathbb{C}$. Consider first an analytic function f in Ω . A well-known result says that if the zero set

$$Z(f) = \{z \in \Omega : f(z) = 0\}$$

of f has a limit point in Ω , then $f(z) = 0$ for all $z \in \Omega$ (see for instance Conway [1, Theorem IV.3.7] or Gamelin [2, Section V.7]). This basic result is often referred to as uniqueness of analytic continuation and has a multitude of applications. Conversely, if E is a discrete subset of Ω with no limit point in Ω , then a famous result of Weierstrass ensures existence of an analytic function f in Ω with $Z(f) = E$ (see for instance Conway [1, Theorem VII.5.15]). Together these results describe exactly the zero sets of analytic functions in planar domains.

A purpose of the present paper is to initiate some similar investigations in the context of harmonic functions. Recall that a function u is said to be harmonic in an open set Ω if it is twice continuously differentiable in Ω (in symbols $u \in C^2(\Omega)$) and satisfies the partial differential equation

$$\Delta u = 0 \quad \text{in } \Omega,$$

where $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2$ is the usual Laplacian. Here $z = x + iy$ is the usual decomposition of a complex number $z \in \mathbb{C}$ into its real and imaginary parts $\operatorname{Re}(z) = x \in \mathbb{R}$ and $\operatorname{Im}(z) = y \in \mathbb{R}$, respectively, and the real line $\mathbb{R}$ is considered a subset of the complex plane $\mathbb{C}$ in the usual way. It is a non-trivial course result that harmonic functions are infinitely differentiable: $u \in C^\infty(\Omega)$ if u is harmonic in Ω (see Conway [1, Theorem III.2.30] and [1, Corollary IV.2.12]).

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We shall need also the standard complex partial derivatives defined by

$$\partial_z = \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \quad \text{and} \quad \bar{\partial}_z = \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right),$$

respectively, where $i^2 = -1$ is the imaginary unit and $z = x + iy$ is as above (see for instance Gamelin [2, Section IV.8]). It is an elementary fact that the Laplacian factorizes as $\Delta = 4\partial\bar{\partial}$ using the differential operators ∂ and $\bar{\partial}$.

We point out that the differential operator $\partial + \bar{\partial}$ is the partial derivative with respect to the real component often denoted $\partial/\partial x$. We can think of $\partial + \bar{\partial}$ as a directional derivative or vector field in the direction of the real axis.

Consider now a complex-valued harmonic function in a planar open set Ω intersecting the real line $\mathbb{R}$. Assume that there exists a point $a \in \Omega \cap \mathbb{R}$ such that

$$(1.1) \quad (\partial + \bar{\partial})^k u(a) = 0 \quad \text{for } k = 0, 1, 2, \dots$$

We prove that then $u(x) = 0$ for every point x in the connected component of $\Omega \cap \mathbb{R}$ containing the point a (see Theorem 4).

Condition (1.1) is satisfied if u vanishes in an interval containing the point a . The example

$$u(z) = \text{Im}(z), \quad z \in \mathbb{C},$$

shows that condition (1.1) is not demanding enough to ensure vanishing of u outside of the real axis. Furthermore, the set $\Omega \cap \mathbb{R}$ can have several connected components and u can be non-vanishing in components of $\Omega \cap \mathbb{R}$ not containing the point a . An example of this phenomenon is furnished by the argument function $u = \arg(\cdot)$ defined in a slit plane, say,

$$\Omega = \{z \in \mathbb{C} : \text{Re}(z) \neq 0 \text{ or } \text{Im}(z) > 0\},$$

where one has $u(x) = 0$ for $x > 0$ and $u(x) = \pi$ for $x < 0$. For reasons indicated above we consider the analysis of condition (1.1) a main contribution of this paper.

As an application of condition (1.1) we prove a uniqueness theorem for harmonic functions involving circular arcs. Let Ω be an open and connected planar set and let C be a suitably curved circular arc contained in Ω . We prove that if u is a complex-valued harmonic function in Ω such that $u(z) = 0$ for $z \in C$, then $u(z) = 0$ for all $z \in \Omega$ (see Theorem 5). As a consequence, if u is an entire harmonic function and u vanishes on a circular arc, then u vanishes identically (see Corollary 3).

This paper is based on joint work with Anders Olofsson. For further information we refer to the paper Khavinson and Shapiro [3] which has also served as a good inspiration. We have merely supplied some details. We mention also a related bachelor's thesis Wright [7].

2. VANISHING OF HARMONIC FUNCTIONS IN THE OPEN UNIT DISC

We analyze first the vanishing property of harmonic functions in simpler domains, which is in the open unit disc.

Let $\mathbb{D}$ be the open unit disc in the complex plane. A well known result [5, Section II] says that a function u is harmonic in $\mathbb{D}$ if and only if it can be expressed as a power series of the form

$$(2.1) \quad u(z) = \sum_{k=0}^{\infty} c_k z^k + \sum_{k=1}^{\infty} c_{-k} \bar{z}^k, \quad z \in \mathbb{D},$$

for some sequence $\{c_k\}_{k=-\infty}^{\infty}$ of complex numbers such that

$$\limsup_{|k| \rightarrow \infty} |c_k|^{1/|k|} \leq 1.$$

Theorem 1. *Let u be a harmonic function in $\mathbb{D}$. Assume that $u(x) = 0$ for $0 < x < a$, where $0 < a < 1$. Then $u(x) = 0$ for $-1 < x < 1$.*

Proof. Recall the power series expansion (2.1). Restricting (2.1) we have that

$$(2.2) \quad u(x) = \sum_{k=0}^{\infty} c_k x^k + \sum_{k=1}^{\infty} c_{-k} x^k = c_0 + \sum_{k=1}^{\infty} (c_k + c_{-k}) x^k$$

for $-1 < x < 1$. By assumption we have that

$$c_0 + \sum_{k=1}^{\infty} (c_k + c_{-k}) x^k = 0$$

for $0 < x < a$. Passing to the limit we have that $\lim_{x \rightarrow 0+} u(x) = c_0 = 0$. A differentiation shows that

$$\sum_{k=0}^{\infty} (k+1) (c_{k+1} + c_{-(k+1)}) x^k = 0$$

for $0 < x < a$. By another passage to the limit as $x \rightarrow 0_+$ we have that $c_1 + c_{-1} = 0$. A further analysis along these lines gives that $c_k + c_{-k} = 0$ for $k = 1, 2, \dots$

We now return to the restricted power series (2.2). From the result of the previous paragraph have that

$$u(x) = c_0 + \sum_{k=1}^{\infty} (c_k + c_{-k})x^k = 0$$

for $-1 < x < 1$ since all the coefficients vanish. This completes the proof of the theorem. $\square$

We point out the next Corollary by applying compositions to u .

Corollary 1. *Let u be a harmonic function in $\mathbb{D}$. Assume that $u(x) = 0$ for $-a < x < 0$, where $0 < a < 1$. Then $u(x) = 0$ for $-1 < x < 1$.*

Proof. Consider the function v defined by $v(z) = u(-z)$ for $z \in \mathbb{D}$. Clearly, the function v is harmonic in $\mathbb{D}$. By assumption we have that $v(x) = 0$ for $0 < x < a$. By Theorem 1 we have that $v(x) = 0$ for $-1 < x < 1$. We conclude that $u(x) = 0$ for $-1 < x < 1$. $\square$

Let $\text{Aut}(\mathbb{D})$ be the set of all one-to-one analytic functions in $\mathbb{D}$ that map $\mathbb{D}$ onto itself. The set $\text{Aut}(\mathbb{D})$ is a group under composition of functions. The object $\text{Aut}(\mathbb{D})$ is known as the automorphism group of $\mathbb{D}$. An interesting type of functions in $\text{Aut}(\mathbb{D})$ is the Möbius transformation (fractional linear transform):

$$(2.3) \quad \varphi_{\alpha}(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}, \quad z \in \mathbb{D},$$

where $\alpha \in \mathbb{D}$. It is well-known that $\varphi \in \text{Aut}(\mathbb{D})$ if and only if it has the form

$$\varphi(z) = e^{i\theta} \varphi_{\alpha}(z), \quad z \in \mathbb{D},$$

where $e^{i\theta} \in \mathbb{T}$ and $\alpha \in \mathbb{D}$ (see [2, Section IX.2]).

We next summarize the properties of $\varphi_{\alpha}(z)$ map.

Proposition 1. [1, Section VI.2] *Let $\varphi_{\alpha}(z)$ in $\mathbb{D}$ for some $\alpha \in \mathbb{D}$, then $\varphi_{\alpha}(z)$ maps $\mathbb{D}$ one-to-one onto itself. Furthermore, it has the following properties:*

- 1) $1 - |\varphi_{\alpha}(z)|^2 = \frac{(1 - |\alpha|^2)(1 - |z|^2)}{(1 - \bar{\alpha}z)^2}$
- 2) $\varphi_{\alpha} \circ \varphi_{\alpha}(z) = z, \forall z \in \mathbb{D}$
- 3) $\varphi'_{\alpha}(z) = -\frac{1 - |\alpha|^2}{(1 - \bar{\alpha}z)^2}$

Proposition 1 is proved along the lines in Conway [1, Section VI.2].

For real numbers a and b with $a < b$ we use the notation (a, b) for the interval

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

When appropriate we consider (a, b) as a subset of the complex plane.

Lemma 1. *Let φ_α be as in (2.3) for some $\alpha \in (-1, 1)$. Then*

$$\varphi_\alpha((a, b)) = (\varphi_\alpha(b), \varphi_\alpha(a)),$$

where $-1 < a < b < 1$.

Proof. Clearly $\varphi_\alpha(\mathbb{R} \cup \{\infty\}) \subset \mathbb{R} \cup \{\infty\}$. From a differentiation formula we have that

$$\varphi'_\alpha(x) = -(1 - \alpha^2)/(1 - \alpha x)^2 < 0$$

for $-1 < x < 1$. By a standard result we have that the restriction of φ_α to the interval $(-1, 1)$ is (strictly) decreasing. By monotonicity we conclude that

$$\varphi_\alpha((a, b)) \subset (\varphi_\alpha(b), \varphi_\alpha(a)).$$

We next prove that $\varphi_\alpha((a, b)) = (\varphi_\alpha(b), \varphi_\alpha(a))$. For this purpose we shall use the involutive character of the map φ_α . Let $y \in (\varphi_\alpha(b), \varphi_\alpha(a))$. Notice that $\varphi_\alpha(\varphi_\alpha(y)) = y$. By the result of the previous paragraph we have that

$$\varphi_\alpha(y) \in (\varphi_\alpha(\varphi_\alpha(a)), \varphi_\alpha(\varphi_\alpha(b))) = (a, b),$$

where we have again used the involutive character of φ_α . Therefore $y \in \varphi_\alpha((a, b))$. This proves that $\varphi_\alpha((a, b)) = (\varphi_\alpha(b), \varphi_\alpha(a))$. $\square$

Theorem 2. *Let u be a harmonic function in $\mathbb{D}$. Assume that $u(x) = 0$ for $a < x < b$, where $-1 < a < b < 1$. Then $u(x) = 0$ for $-1 < x < 1$.*

Proof. Recall the automorphism map (2.3) and consider the composition $v = u \circ \varphi_b$. Clearly v is harmonic in $\mathbb{D}$. Set

$$c = \varphi_b(a) = (b - a)/(1 - ab) \in (0, 1).$$

By Lemma 1 we have that

$$\varphi_b((0, c)) = (\varphi_b(c), b) = (a, b),$$

where in the last step we have used the involutive property of φ_α . By assumption we have that $v(x) = 0$ for $0 < x < c$. Therefore by Theorem 1 we have that $v(x) = 0$ for $-1 < x < 1$. We now return to the function u and conclude that $u(x) = 0$ for $-1 < x < 1$. This completes the proof of Theorem. $\square$

3. PRELIMINARIES

We point out some important preliminaries regarding concepts in topology. We denote by

$$B(a, r) = \{z \in \mathbb{C} : |z - a| < r\}$$

the open disc with center $a \in \mathbb{C}$ and radius $r > 0$. An open planar set Ω is a union of discs. A set F is closed in $\mathbb{C}$ if its complement $\mathbb{C} \setminus F$ is open in $\mathbb{C}$. In this way the complex plane $\mathbb{C}$ is a metric space defined by the metric

$$d(z_1, z_2) = |z_1 - z_2|, \quad z_1, z_2 \in \mathbb{C}.$$

A subset E of $\mathbb{C}$ is considered a metric space on its own right by restricting the above metric d to the set $E \times E$. This topologization of E is often referred to as the relative topology.

A metric space X is called connected if it can not be represented as a disjoint union of two non-empty open subsets. A connected component of X is a maximal connected subset of X , that is, a subset X_0 of X is a connected component of X if it is connected and has the property that $X_0 \subset E \subset X$ and E connected implies that $E = X_0$. Since every singleton set is connected, we deduce that connected components are non-empty. A fundamental result in the theory of connectedness says that every topological space can be decomposed as a disjoint union of connected components (see Theorem C in Section 32 in [6]).

Let us discuss connected subsets of $\mathbb{R}$. Recall that a subset E of $\mathbb{C}$ is called convex if for every $z_1, z_2 \in E$ the line segment

$$[z_1, z_2] = \{(1 - t)z_1 + tz_2 : 0 \leq t \leq 1\}$$

is contained in E . A linear combination of the form $(1 - t)z_1 + tz_2$ with $0 \leq t \leq 1$ is commonly called a convex combination of z_1 and z_2 .

Proposition 2. *A subset E of $\mathbb{R}$ is connected if and only if it is convex.*

Proof. We first consider the only if part. Assume E is connected but not convex, then there exists $x, y \in E$, $x < y$ such that there is another point $z \notin E$ with $x < z < y$. Then there is a disconnection of $E = (E \cap (-\infty, z)) \cup (E \cap (z, +\infty))$. Therefore E is not connected, which is a contradiction, hence E is convex.

We next consider the if part. Assume E is a convex set. Then for every $x, y \in E$, $x < y$ we have the interval $[x, y] \subseteq E$. Assume E is not connected. Then there exist U, V disjoint non-empty open sets with $E = U \cup V$. Let points $u \in U$ and $v \in V$, $u < v$ then $[u, v] \subseteq E$, by convexity. Let $z = \sup([u, v] \cap U)$, by openness of U and V , we have

$z \notin U$ and $z \notin V$. This contradicts the fact that $z \in E$, hence E is connected. This concludes the proof. $\square$

We point out that the least upper bound property of real numbers is used in an essential way in the above proof.

Let $E \subset \mathbb{R}$. We next define a relation $\sim$ on the set E . We say that $x_1 \in E$ and $x_2 \in E$ are related $x_1 \sim x_2$ if the interval

$$[\min(x_1, x_2), \max(x_1, x_2)]$$

is contained in E . In view of the linear order structure of $\mathbb{R}$ this relation $\sim$ is well-defined. Notice that $x_1 \sim x_2$ means that the interval with endpoints x_1 and x_2 is contained in E .

Proposition 3. *Let $E \subset \mathbb{R}$ and consider the relation $\sim$ defined above. Then the relation $\sim$ is an equivalence relation on E . Furthermore, the partition of E induced by $\sim$ is the decomposition of E into connected components.*

Proof. We first show that the relation $\sim$ is an equivalence relation on E . Clearly, the relation $\sim$ is reflexive and symmetric. We proceed to prove the transitivity property of $\sim$. Assume that $a \sim b$ and $b \sim c$. Then

$$[\min(a, b), \max(a, b)] \subseteq E \quad \text{and} \quad [\min(b, c), \max(b, c)] \subseteq E.$$

Since $\min(b, c) \leq b \leq \max(a, b)$ and $\min(a, b) \leq b \leq \max(b, c)$, we have that

$$[\min(a, b, c), \max(a, b, c)] \subseteq E.$$

As a consequence $[\min(a, c), \max(a, c)] \subseteq E$. Therefore $a \sim c$. This proves transitivity of $\sim$.

The equivalence relation $\sim$ induces a partition of E . We next consider the equivalence class

$$C(x_0) = \{x \in E : x \sim x_0\}$$

of an element $x_0 \in E$.

We first prove that $C(x_0)$ is connected. Let $x_1, x_2 \in C(x_0)$. Without loss of generality we can assume that $x_1 \leq x_2$. Let

$$x = (1 - t)x_1 + tx_2$$

be a convex combination of x_1 and x_2 . We shall prove that $x \in C(x_0)$. Since $x_1, x_2 \in C(x_0)$ we have that $x_1 \sim x_0$ and $x_2 \sim x_0$. By symmetry and transitivity we have that $x_1 \sim x_2$. In view of definition of $\sim$ this means that the interval $[x_1, x_2]$ with end points x_1 and x_2 is contained in E . Since $x \in [x_1, x_2]$ we have that $x \in E$ and $[x_1, x] \subset E$. Therefore $x_1 \sim x$. Since also $x_0 \sim x_1$ we have from transitivity that $x_0 \sim x$.

Therefore $x \in C(x_0)$. This proves that every convex combination of x_1 and x_2 belongs to $C(x_0)$. Varying $x_1, x_2 \in C(x_0)$ we have that $C(x_0)$ is convex. By Proposition 2 we have that $C(x_0)$ is connected.

We next prove that $C(x_0)$ is a connected component of E . Assume that $C(x_0) \subset D \subset E$ and D is connected. We prove that $D = C(x_0)$. Let $x \in D$ and observe that D is convex by Proposition 2. Since also $x_0 \in D$ by assumption, we have that the interval with end points x_0 and x is contained in D . Therefore $x_0 \sim x$ and $x \in C(x_0)$. This proves that $D = C(x_0)$.

We next prove that every connected component of E has the form $C(x_0)$ for some $x_0 \in E$. Let D be such a connected component of E . Since D is non-empty there exists $x_0 \in D$. We first prove that $D \subset C(x_0)$. Let $x \in D$. By Proposition 2 the set D is convex so the interval with endpoints x_0 and x is contained in D . Therefore $x_0 \sim x$ and $x \in C(x_0)$. This proves that $D \subset C(x_0)$. Since $C(x_0)$ is connected, we conclude from the maximality property of D that $D = C(x_0)$. $\square$

Notice that the non-empty convex subsets of the set $\mathbb{Q}$ of rational numbers are the singleton subsets. We conclude from Proposition 2 that the connected components of $\mathbb{Q}$ are its singleton subsets. Such a space is sometimes called totally disconnected.

A well-known result says that a function u in a disc $B(a, r)$ is harmonic in $B(a, r)$ if and only if it has a power series expansion of the form

$$(3.1) \quad u(z) = \sum_{k=0}^{\infty} c_k (z - a)^k + \sum_{k=1}^{\infty} c_{-k} (\bar{z} - \bar{a})^k, \quad z \in B(a, r),$$

for some sequence $\{c_k\}_{k=-\infty}^{\infty}$ of complex numbers such that

$$(3.2) \quad \limsup_{|k| \rightarrow \infty} |c_k|^{1/|k|} \leq 1/r.$$

In fact, this follows by a corresponding expansion result for analytic functions combined with existence of a harmonic conjugate. See Conway [1, Theorem IV.2.8] and [1, Theorem III.2.30] for details.

Recall that convergent power series expansions are uniformly convergent on compact subsets of its disc of convergence (see Conway [1, Theorem III.1.3]). Recall also that convergent power series expansions are differentiated termwise (see Conway [1, Proposition III.2.5]). As a consequence of (3.1) we have that $u \in C^\infty(\Omega)$ if u is harmonic in an open set Ω . For similar reason we have that $c_0 = u(a)$,

$$(3.3) \quad c_k = \frac{\partial^k u(a)}{k!} \quad \text{and} \quad c_{-k} = \frac{\bar{\partial}^k u(a)}{k!}$$

for $k \geq 1$, where the c_k 's are as in (3.1).

The following uniqueness result is well-known.

Theorem 3. *Let Ω be an open and connected planar set and $a \in \Omega$. Let u be a complex-valued harmonic function in Ω . Assume that $u(a) = 0$ and*

$$\partial^k u(a) = \bar{\partial}^k u(a) = 0$$

for $k = 1, 2, \dots$. Then $u(z) = 0$ for $z \in \Omega$.

Theorem 3 is proved along similar lines as the corresponding result for analytic functions (see Gamelin [2, Section V.4]). We point out that the series expansion (3.1) can be used as a replacement for the standard power series expansion of an analytic function.

We denote by $\mathbb{Z}$ the set of integers.

Proposition 4. *Let u be a complex-valued harmonic function in $B(a, r)$ of the form (3.1) for some sequence $\{c_k\}_{k=-\infty}^{\infty}$ of complex numbers satisfying (3.2). Then*

$$(3.4) \quad c_k = \frac{1}{2\pi\rho^{|k|}} \int_{-\pi}^{\pi} u(a + \rho e^{i\theta}) e^{-ik\theta} d\theta$$

for $k \in \mathbb{Z}$, where $0 < \rho < r$.

Proof. From the power series expansion (3.1) we have that

$$u(a + \rho e^{i\theta}) = \sum_{m=0}^{\infty} c_m \rho^m e^{im\theta} + \sum_{m=1}^{\infty} c_{-m} \rho^m e^{-im\theta}.$$

Integrating termwise we have that

$$\begin{aligned} \int_{-\pi}^{\pi} u(a + \rho e^{i\theta}) e^{-ik\theta} d\theta &= \sum_{m=0}^{\infty} c_m \rho^m \int_{-\pi}^{\pi} e^{i(m-k)\theta} d\theta \\ &\quad + \sum_{m=1}^{\infty} c_{-m} \rho^m \int_{-\pi}^{\pi} e^{-i(m+k)\theta} d\theta. \end{aligned}$$

Notice that the change of order of summation and integration is permitted by uniform convergence. From orthogonality of exponential monomials we have that

$$\int_{-\pi}^{\pi} u(a + \rho e^{i\theta}) e^{-ik\theta} d\theta = c_k \rho^{|k|} 2\pi.$$

This completes the proof of the proposition. $\square$

From Proposition 4, it is straightforward to check that the coefficient integral (3.4) can be represented as a line integral by a change of variables $z \rightarrow a + \rho e^{i\theta}$:

$$c_k = \frac{1}{2\pi i} \oint_{|z-a|=\rho} \frac{u(z)}{(z-a)^{k+1}} dz \text{ and } c_{-k} = -\frac{1}{2\pi i} \oint_{|z-a|=\rho} \frac{u(z)}{(\bar{z}-a)^{k+1}} d\bar{z},$$

for $k \in \mathbb{N}$ and $0 < \rho < r$, where the integration is traversed in the positive orientation.

4. VANISHING CONDITIONS OF HARMONIC FUNCTIONS IN GENERAL DOMAINS

Recall that a function u is said to be harmonic in Ω if $u \in C^2(\Omega)$ and $\partial\bar{\partial}u = 0$ in Ω , where

$$\partial = \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \quad \text{and} \quad \bar{\partial} = \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right)$$

are the standard complex partial derivatives.

Lemma 2. *Let u be a complex-valued harmonic function in a planar open set Ω . Let $k \in \mathbb{Z}^+$. Then $(\partial + \bar{\partial})^k u = \partial^k u + \bar{\partial}^k u$ in Ω .*

Proof. By the binomial theorem we know that

$$(\partial + \bar{\partial})^k = \sum_{j=0}^k \binom{k}{j} \partial^j \bar{\partial}^{k-j},$$

where we have used equality of mixed partial derivatives. Since u is harmonic we have that $\partial\bar{\partial}u = 0$ in Ω . Differentiating we have that

$$(\partial + \bar{\partial})^k u = \sum_{j=0}^k \binom{k}{j} \partial^j \bar{\partial}^{k-j} u = \partial^k u + \bar{\partial}^k u$$

in Ω . □

Observe that the differential operator $\partial + \bar{\partial} = \partial/\partial x$ appearing above is the partial derivative on the real part of the complex variable $z = x + iy$.

Theorem 4. *Let u be a complex-valued harmonic function in a planar open set Ω . Let $a \in \Omega \cap \mathbb{R}$. Assume that $u(a) = 0$ and*

$$\partial^k u(a) + \bar{\partial}^k u(a) = 0$$

for $k = 1, 2, \dots$. Then $u(x) = 0$ for every point x in the connected component of $\Omega \cap \mathbb{R}$ containing a .

Proof. For simplicity of notation we set $X = \Omega \cap \mathbb{R}$, which is an open subset of $\mathbb{R}$. Consider the set

$$E = \left\{ x \in X : u(x) = 0 \text{ and } \partial^k u(x) + \bar{\partial}^k u(x) = 0 \text{ for } k = 1, 2, \dots \right\}.$$

Clearly $E \subset X$. We shall prove that E is an open and closed subset of X .

We first prove that E is a closed subset of X . Observe that

$$X \setminus E = \{x \in X : u(x) \neq 0\} \cup \bigcup_{k=1}^{\infty} \{x \in X : \partial^k u(x) + \bar{\partial}^k u(x) \neq 0\}.$$

The set $X \setminus E$ is open in X since it is a union of open subsets of X . This proves that E is a closed subset of X .

We next prove that E is an open subset of X . Let $x_0 \in E$. Since $x_0 \in \Omega$ and Ω is an open planar set, there exists $r > 0$ such that $B(x_0, r) \subset \Omega$. Since u is harmonic in $B(x_0, r)$ it has a series expansion

$$u(z) = \sum_{k=0}^{\infty} c_k (z - x_0)^k + \sum_{k=1}^{\infty} c_{-k} (\bar{z} - x_0)^k$$

convergent for $z \in B(x_0, r)$. By coefficient formulas (3.3) we have that $c_0 = u(x_0)$, $c_k = \partial^k u(x_0)/k!$ and $c_{-k} = \bar{\partial}^k u(x_0)/k!$ for $k \geq 1$. Since $x_0 \in E$ we have that $c_0 = 0$ and $c_k + c_{-k} = 0$ for $k \geq 1$. For $x_0 - r < x < x_0 + r$ we now have that

$$\begin{aligned} u(x) &= \sum_{k=0}^{\infty} c_k (x - x_0)^k + \sum_{k=1}^{\infty} c_{-k} (x - x_0)^k \\ &= c_0 + \sum_{k=1}^{\infty} (c_k + c_{-k}) (x - x_0)^k = 0 \end{aligned}$$

since the coefficients all vanish. From here we conclude that $u(x) = 0$ for $x \in (x_0 - r, x_0 + r)$. Let $k \in \mathbb{Z}^+$. Recall that the differential operator $\partial + \bar{\partial}$ has an interpretation of partial derivative on the real part of the complex variable. Differentiating we have that

$$(\partial + \bar{\partial})^k u(x) = 0$$

for $x \in (x_0 - r, x_0 + r)$. By Lemma 2 we have that $\partial^k u(x) + \bar{\partial}^k u(x) = 0$ for $x \in (x_0 - r, x_0 + r)$. In this way we conclude that $(x_0 - r, x_0 + r) \subset E$. Therefore E is open in X .

We now deduce the conclusion of the theorem. Since the set X is open in $\mathbb{R}$, it has the form of a disjoint union of at most countably many disjoint open convex intervals, this is a similar result of [1, Theorem 2.9]. These intervals are the connected components of X . Since the set E is open and closed in X , the set E is a union of connected components

of X [1, Theorem 2.7]. Since $a \in E$ by assumption, we conclude that the connected component of X containing the point a is a subset of E . This yields the conclusion of the theorem. $\square$

Remark. We point out that the vanishing condition in Theorem 4 can be formulated saying that

$$(\partial + \bar{\partial})^k u(a) = 0 \quad \text{for } k = 0, 1, 2, \dots$$

In fact, this follows from Lemma 2.

We next discuss harmonic functions vanishing on an interval.

Corollary 2. *Let u be a complex-valued harmonic function in a planar open set Ω . Let $a, b \in \Omega \cap \mathbb{R}$ be such that $a < b$. Assume that $u(x) = 0$ for $a < x < b$. Then $u(x) = 0$ for every point x in the connected component of $\Omega \cap \mathbb{R}$ containing the interval (a, b) .*

Proof. Take $x_0 \in (a, b)$. By assumption we have $u(x_0) = 0$. A differentiation shows that

$$(\partial + \bar{\partial})^k u(x_0) = 0.$$

By Lemma 2 we have

$$\partial^k u(x_0) + \bar{\partial}^k u(x_0) = 0,$$

for $k = 1, 2, 3, \dots$. By Theorem 4 we conclude that $u(x) = 0$ for x in the connected component of $\Omega \cap \mathbb{R}$ containing the interval (a, b) . $\square$

We next consider functions vanishing on a circular arc.

Theorem 5. *Let Ω be an open and connected planar set. Let $a \in \Omega$ and assume that the disc $B(a, r)$ is such that the closed disc $\overline{B(a, r)}$ is contained in Ω , where $r > 0$. Let C be a subarc of positive length of the circle $\partial B(a, r)$. Let u be a complex-valued harmonic function in Ω and assume that $u(z) = 0$ for $z \in C$. Then $u(z) = 0$ for $z \in \Omega$.*

Proof. The Möbius transformation

$$f(z) = \frac{z - i}{z + i}$$

maps the extended real line $\mathbb{R} \cup \{\infty\}$ onto the unit circle $\mathbb{T}$ and takes the point ∞ to the point 1. We set

$$T(z) = r \frac{z - i}{z + i} + a.$$

For similar reason the Möbius transformation T maps the real line onto the set $\partial B(a, r) \setminus \{a + r\}$. Consider now the composition

$$v(z) = u(T(z))$$

defined for $z \in T^{-1}(\Omega)$. Clearly, $v(z)$ is harmonic for $z \in T^{-1}(\Omega)$. Observe that $\mathbb{R}$ is contained in the set $T^{-1}(\Omega)$. By assumption we see that the function v vanishes on an interval. By Corollary 2 we have that $v(x) = 0$ for $x \in \mathbb{R}$. We now return to the function u and conclude that $u(z) = 0$ for $z \in \partial B(a, r) \setminus \{a + r\}$. By continuity we also have that $u(a + r) = 0$. By Proposition 4 we have that $u(z) = 0$ for $z \in B(a, r)$. Finally by Theorem 3 we conclude that $u(z) = 0$ for $z \in \Omega$. $\square$

By an entire harmonic function we understand a function which is harmonic in the whole complex plane.

Corollary 3. *Let u be a complex-valued entire harmonic function. Assume that u vanishes on a circular arc. Then $u(z) = 0$ for all $z \in \mathbb{C}$.*

Corollary 3 follows from Theorem 5 by choosing $\Omega = \mathbb{C}$.

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